

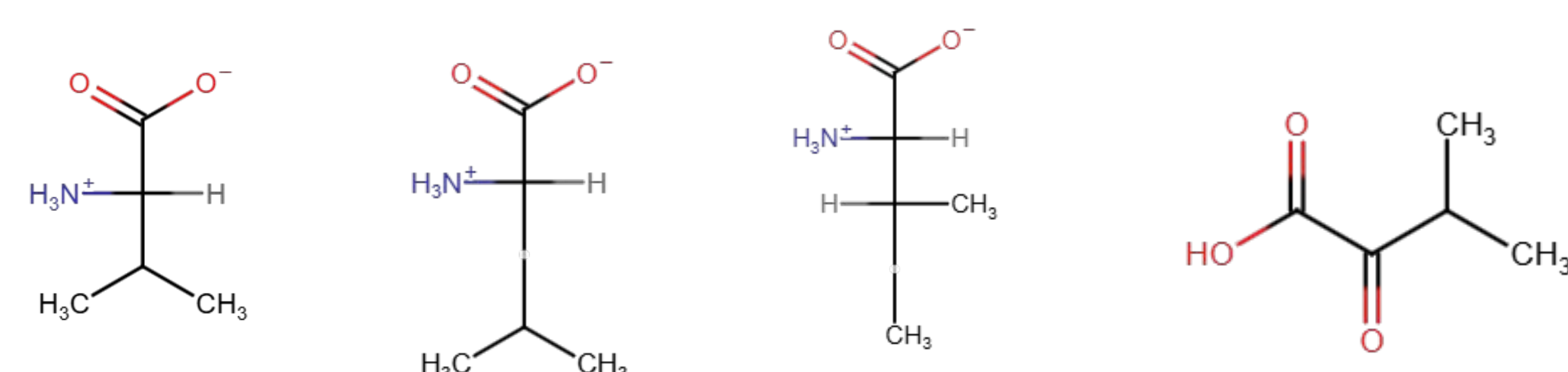
Colorimetric Paperfluidic Assay for MSUD

Detection Through Alpha Keto Acids

Mahale Silva, Marilynn Tirado, Shannon H. Hilton
Office of Undergraduate Research, University of Maryland

INTRODUCTION

Maple syrup urine disease (MSUD) is a rare genetic disorder affecting the metabolism of branched-chain amino acids, causing the buildup of toxic byproducts, branched-chain α -ketoacids, in the body.

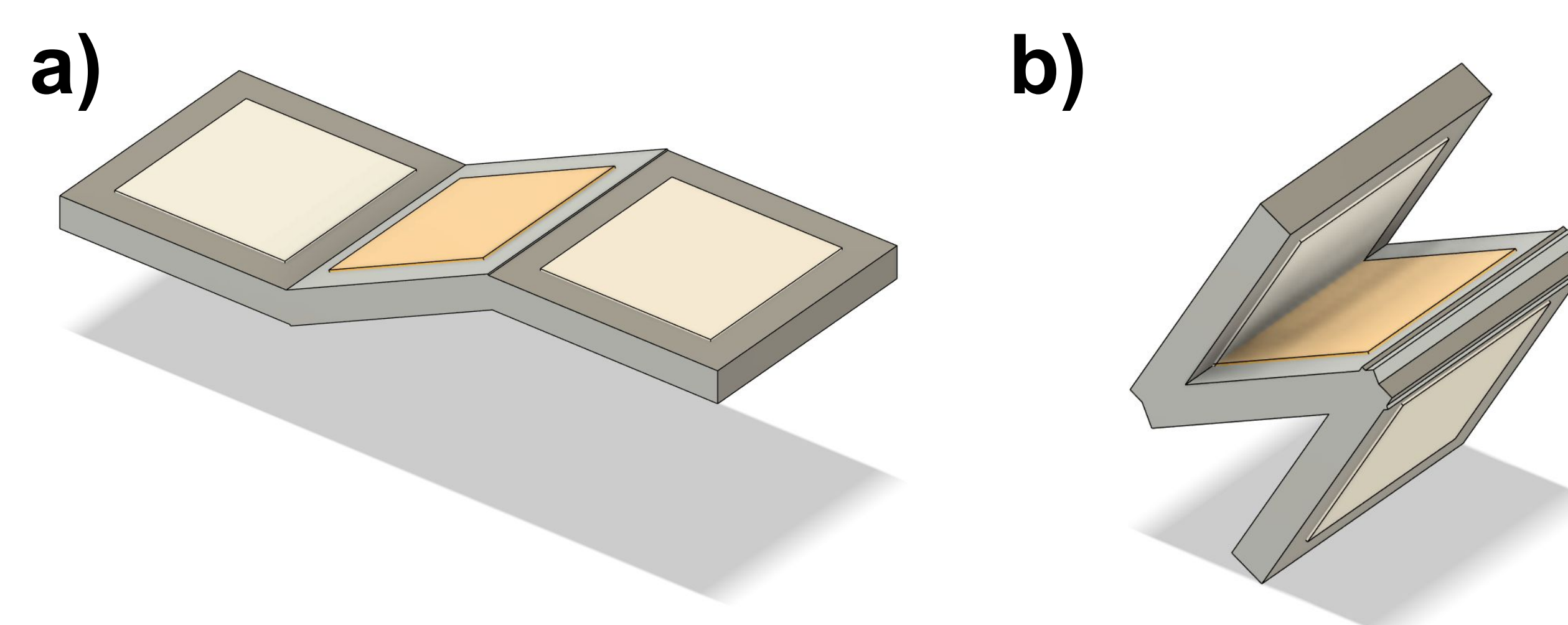


Left to right: 3 branched-chain amino acids valine, leucine, and isoleucine, and KIV

Symptoms of classical MSUD, the most severe form, include delayed developmental milestones, lethargy, difficulty feeding, and sweet-smelling urine. Left untreated, it leads to irreversible neurological injury within 7 to 10 days after birth, and death occurs within 2 months¹.

Testing for MSUD consists of a blood or urine screening, which takes time and advanced technology. Our goal is to develop a simpler paperfluidic colorimetric device for the disease, targeting a byproduct of valine, α -ketoisovalerate (KIV).

DEVICE CONCEPT



The origami paperfluidic device proposed has 3 paper spots for samples (shown on both sides for accessibility to middle). Following (a), the left spot contains sodium hydroxide, the middle contains Brady's reagent (DNPH and Sulfuric Acid Combined), and the right contains ninhydrin. KIV sample is added to the center spot, and then ~5-10 minutes should pass until the ninhydrin spot should be folded onto it. Lastly, the sodium hydroxide is folded onto the center spot from the opposite direction, as seen in (b). The device can then be heated to optimize results.

RESULTS/DISCUSSION



Figure 1. Spiking of 0.003 M KIV samples containing 10mM Brady's, 2% Ninhydrin, and 1×10^{-6} M NaOH with KIV concentrations of 14.0 mM, 5.3 mM, 4.0 mM, 1.3 mM, 0.2 mM, and 0 mM.

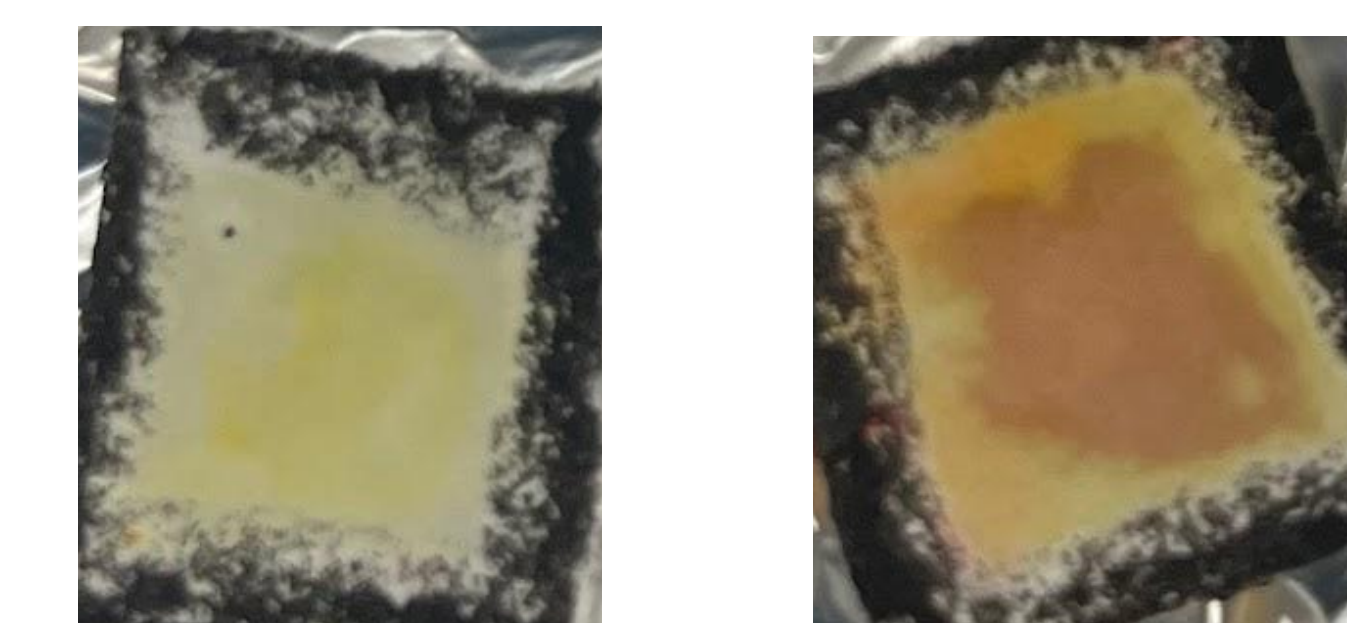
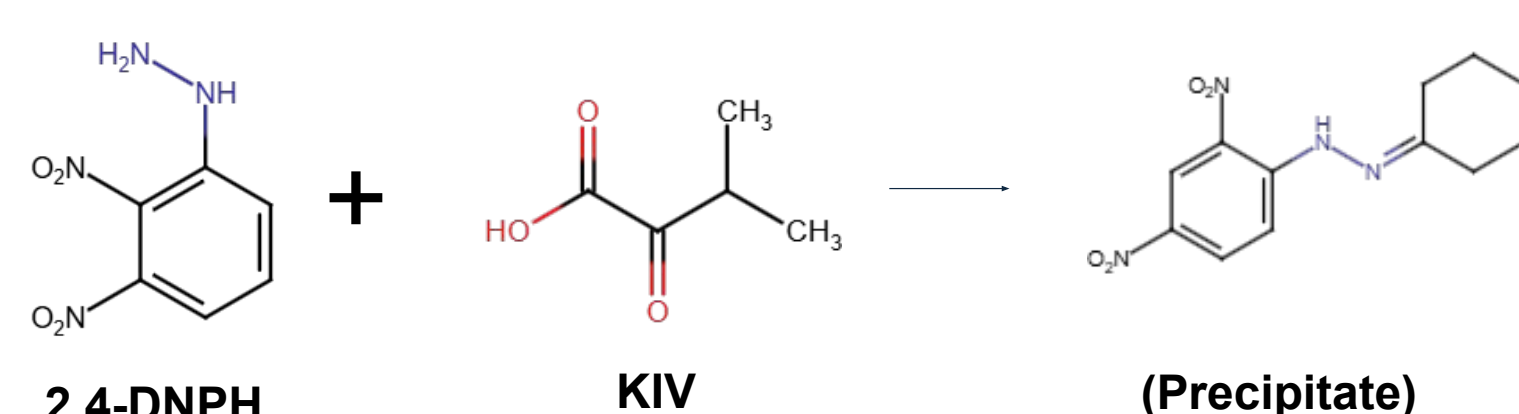


Figure 2. Proof of concept at low concentrations for paper tests. Whatman paper first soaked in 2% Ninhydrin, then has 5 mM Brady's, KIV (4 mM and 0.5 mM left to right), and 67 mM NaOH added in sequence with drying between each step.

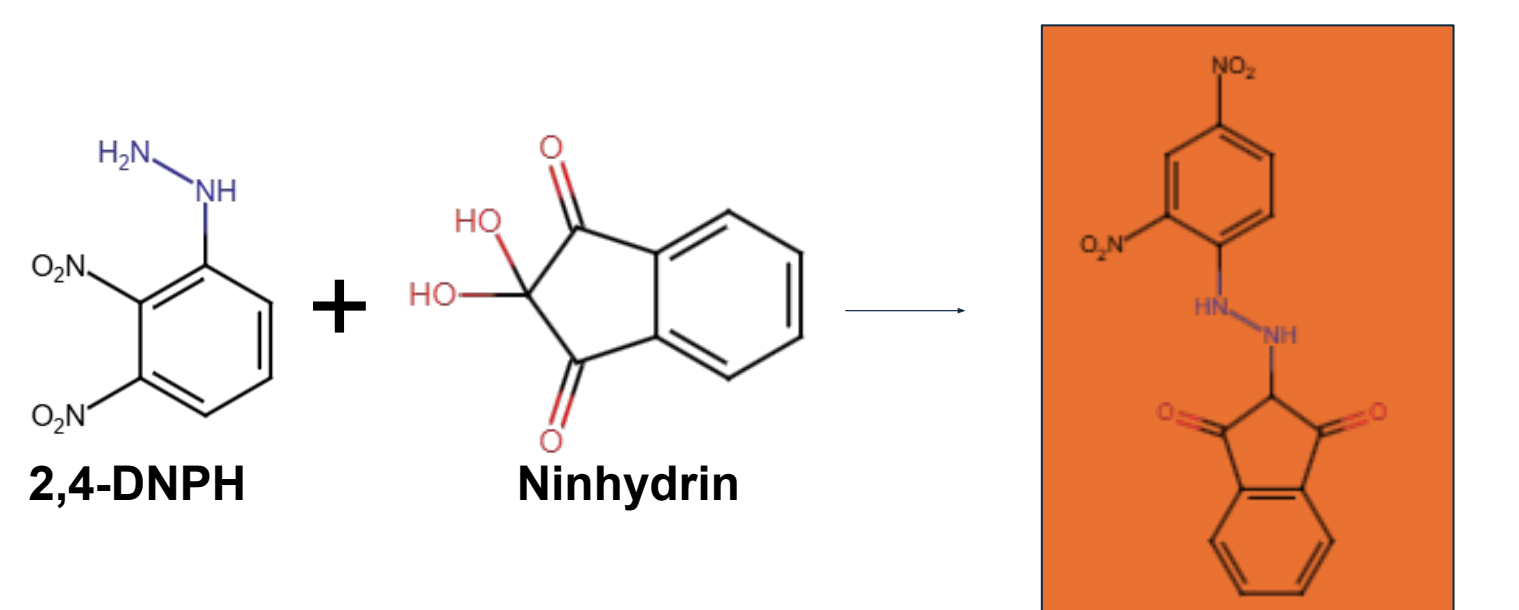
- While lower concentration and tighter ranges of KIV concentrations could not yet be utilized for the origami folding mechanism, it is possible for the proper color changes to show on paper at low and tighter KIV concentrations.

METHODS

2,4-DNPH Test



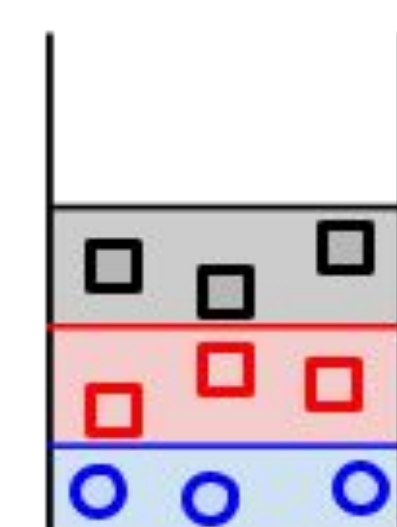
Leftover DNPH Detection²



In Solution:

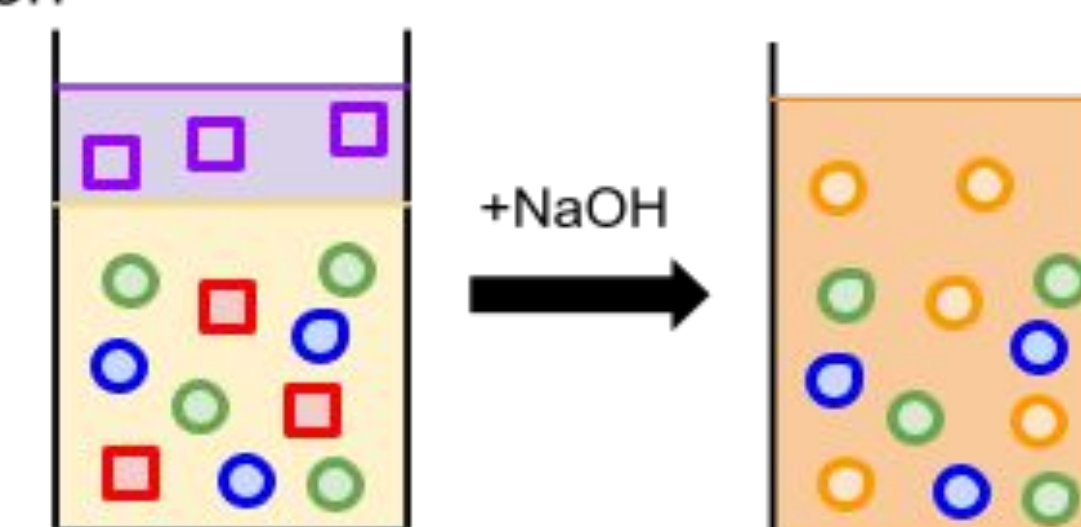
First Reaction

- KIV
- Sulfuric Acid
- DNPH
- Precipitate

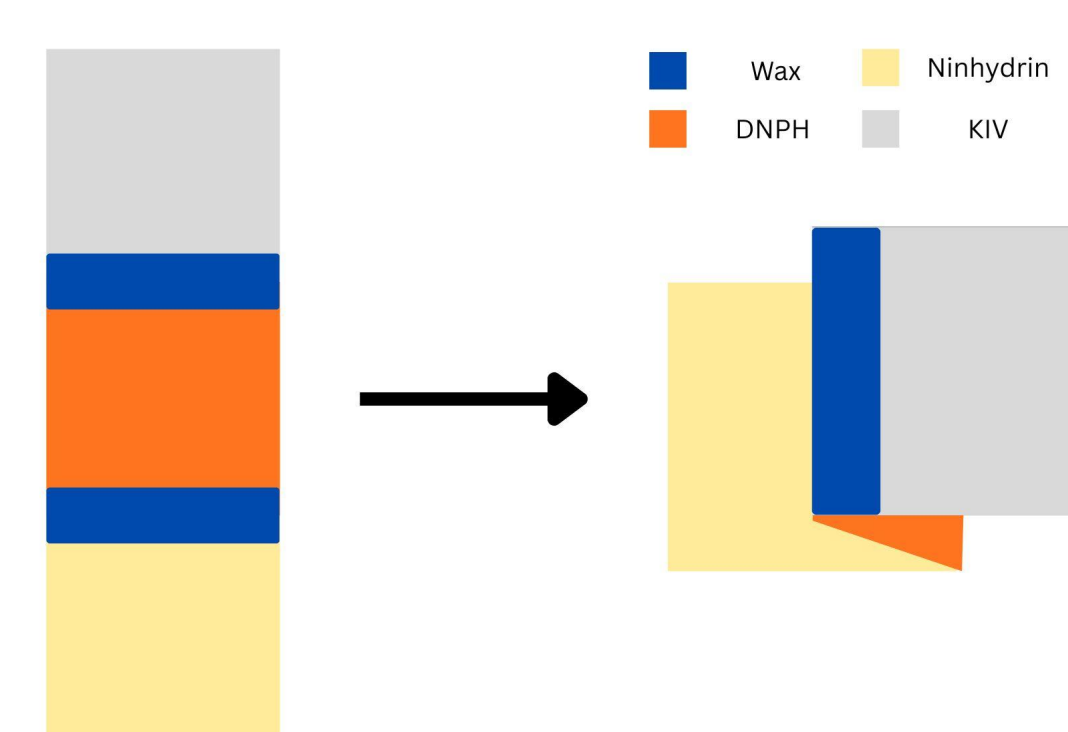


Second Reaction

- Ninhydrin
- Key Product



On Paper:



- Wax barrier is made between each section
- Ninhydrin and DNPH is added to their respective sections
- Empty section is folded over top the middle section and NaOH then KIV is added
- Paper is wrapped in foil and heated

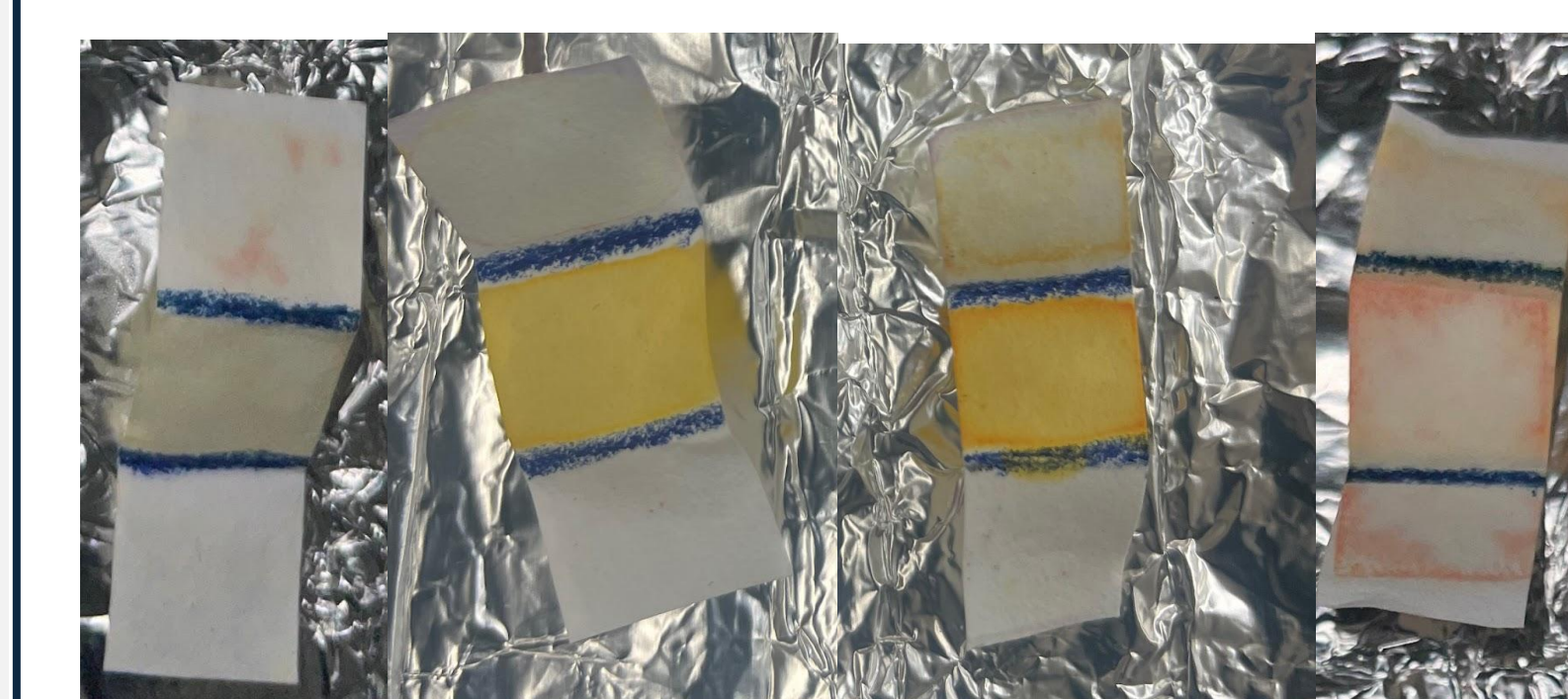


Figure 3. Paper tests using folding mechanism using 10mM Brady's, 2% Ninhydrin, and 1 μ M NaOH with KIV concentrations of 143 mM (pos. control), 10 mM, 1mM, and 0 mM (neg. control).

- Positive and negative controls showed clear and correct differences in color, signifying that the reaction works properly using the mechanism.
- Spiking showed a lower limit of detection in solution, this limit has not yet been tested on paper

NEXT STEPS

- Complete paperfluidic tests using folding mechanism at lower concentrations
- Show a greater color gradient with smaller gaps in concentration
- Develop a functional paperfluidic device and run test trials

ACKNOWLEDGEMENTS

Thank you to the First Year Innovation and Research Experience (FIRE) program at the University of Maryland.

REFERENCES

- Hassan SA, Gupta V. Maple Syrup Urine Disease. [Updated 2024 Mar 3]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK557773/>
- Elgendy, K.; Alaa, M.; Elmosallamy, E.; Hassan, M.; Gamal, H. APPLICATIONS of NINHYDRIN: SPECTROPHOTOMETRIC DETERMINATION of HYDRAZINE and HYDROXYLAMINE (Aplikasi Ninhidrin: Penentuan Spektrofotometri Bagi Hidrazin Dan Hidroksil Amina). Malaysian Journal of Analytical Sciences 2022, 26, 855–866.
- Structures drawn with Marvin JS by Chemaxon.